

Lab record-04

Question 1:

Given a string, count the number of vowels (a, e, i, o, u).

Examples:

Input: "hello"

Output: 2

Input: "education"

Output: 5

Question 2:

Reverse a given string without using library functions.

Examples:

Input: "world"

Output: "dlrow"

Input: "python"

Output: "nohtyp"

Question 3:

Check whether a given string is a palindrome or not.

(A string is palindrome if it reads the same backward as forward.)

Examples:

Input: "madam"

Output: Palindrome

Input: "hello"

Output: Not Palindrome

Question 4:

Count the number of words in a given string.

Examples:

Input: "I love programming"

Output: 3

Input: "ChatGPT is awesome"

Output: 3

Question 5:

Find the frequency of each character in a string.

Examples:

Input: "apple"

Output:

```
a → 1  
p → 2  
l → 1  
e → 1
```

Input: "banana"

Output:

```
b → 1  
a → 3  
n → 2
```

Question 6:

Convert all lowercase letters in a string to uppercase and vice versa.

Examples:

Input: "HeLLo"

Output: "hElLo"

Input: "Python"

Output: "pYTHON"

Question 7:

Find the length of a string without using the built-in `strlen()` function.

Examples:

Input: "OpenAI"

Output: 6

Input: "C programming"

Output: 13

Question 8:

Remove all spaces from a given string.

Examples:

Input: "Hello World"

Output: "HelloWorld"

Input: "Data Structures"

Output: "DataStructures"

Question 9:

Count the number of digits, alphabets, and special characters in a string.

Examples:

Input: "A1@B2#C3"

Output:

Alphabets = 3

Digits = 3

Special Characters = 2

Question 10:

Find the first non-repeating character in a string.

Examples:

Input: "swiss"

Output: "w"

Input: "success"

Output: "u"
